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# Kaan: On-Device Acoustic Screening for Stored-Grain Pests under Phone-Mic Domain Shift

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## Abstract

Post-harvest insect damage is a major food-security and climate-relevant loss pathway in stored grain. We present **Kaan**, an open, offline smartphone system that classifies contact acoustics into clean grain versus three primary stored-rice pests using a compact INT8 CNN. Rather than chasing laboratory accuracy alone, we emphasize deployment constraints: leakage-aware evaluation, multi-approach bake-offs, teacher distillation into a phone-sized model, temperature calibration with selective abstention, and a *robustness ladder* that stress-tests phone-like degradations without new field recordings. On IRRI acoustic data with ambient clean windows, a distilled production model reaches 97.2% validation accuracy; multi-seed bake-offs show classical and deep models both exceed a cited 84.51% reference under our protocol. The robustness ladder reveals sharp drops under additive noise and telephone band-limiting: a limitation we treat as a first-class result for climate-adaptation tooling that must survive rural handsets. Code and models are released under Apache-2.0.

## 1 Climate impact pathway

Reducing avoidable post-harvest loss lowers pressure to expand production and associated emissions, while improving food availability for smallholders IGMRI (2015); World Bank and FAO. Acoustic screening can catch internal feeders before visual symptoms, enabling earlier aeration, sunning, or fumigation decisions. Kaan targets *adaptation* (protecting stored food under climate stress on storage conditions) and indirect *mitigation* (less waste). We deliberately optimize for offline phones: no cloud inference, multilingual advisories, and models small enough for browser/ONNX and TFLite.

## 2 Problem and system

**Classes:** clean, rice weevil (*Sitophilus oryzae*), lesser grain borer (*Rhyzopertha dominica*), red flour beetle (*Tribolium castaneum*). **Pipeline:** 10 s @ 16 kHz mono  $\rightarrow$   $128 \times 128$  mel  $\rightarrow$  INT8 CNN  $\rightarrow$  confidence gate. Pest WAVs come from the IRRI Rice Acoustic Sensor dataset Balingbing et al. (2024); clean uses Speech Commands ambient windows rather than synthetic silence. Splits are stratified at *file* level after byte-deduplication to avoid train/val twins.

## 3 Methods

**Bake-off.** Eleven approaches (mel-CNNs, 1D CNN, YAMNet probe, SVM/MLP/GBDT/RF/ExtraTrees/kNN/logreg) share one leakage-aware split; we report multi-seed means, bootstrap CIs, and McNemar tests.

Table 1: Multi-seed bake-off accuracy (mean $\pm$ std; seeds 42/43/44).

| Approach     | Acc mean $\pm$ std | Seeds > 84.51% |
|--------------|--------------------|----------------|
| GBDT         | 95.36 $\pm$ 1.50   | 3/3            |
| CNN deep     | 95.15 $\pm$ 0.48   | 3/3            |
| ExtraTrees   | 94.94 $\pm$ 1.10   | 3/3            |
| SVM-RBF      | 94.73 $\pm$ 1.20   | 3/3            |
| YAMNet probe | 85.65 $\pm$ 1.83   | 2/3            |
| CNN-ID       | 64.77 $\pm$ 20.7   | 1/3            |

Table 2: Robustness ladder on seed-42 baseline (clean = 97.15%).

| Rung                              | Acc         |
|-----------------------------------|-------------|
| Muffle / reverb / compress / gain | 92–97%      |
| Clip                              | 87.7%       |
| Phone band-pass                   | 52.2%       |
| SNR 20 dB                         | 30.4%       |
| SNR $\leq$ 10 dB / phone combos   | $\sim$ 7.6% |

**Distillation.** Soft labels from GBDT + ExtraTrees + deep CNN train a deep student for deployment (TFLite  $\sim$ 333 KB; ONNX for web).

**Robustness ladder.** Without field mics, we replay validation audio under pink SNR, phone band-pass, muffling, compression, clipping, reverb, gain/AGC, and combinations, reporting accuracy per rung.

**Calibration.** Temperature scaling fitted on a validation slice; abstain curves trade coverage for accuracy.

**Confusion-focused heads.** Cost-sensitive upweighting and a fine-tuned weevil $\leftrightarrow$ borer specialist on the baseline encoder with *strict* fusion (override only when top-2 are that pair and margin is small; gate swept on val).

**SSL.** SimCLR-style SpecAugment views on IRRI+ambient mels, then supervised fine-tune.

## 4 Results

Table 1 summarizes the multi-seed bake-off (reference line: cited 84.51% Balingbing et al. (2024) under *our* protocol, not a locked reimplementation). Distillation (seed 42) yields **97.15%** val accuracy (macro-F1 0.977) versus 95.57% hard-only deep and 95.89% teacher ensemble. SSL fine-tune reaches 97.78% on the same seed. Calibration reduces ECE (e.g., 0.044  $\rightarrow$  0.030 at  $T=0.5$  on seed 42).

Table 2 is the deployment-facing result: laboratory mel-CNNs survive mild muffling but **fail under phone-band and noise**, exactly the shift rural handsets induce. From-scratch hierarchical cascades collapse; fine-tuned specialists reach  $\sim$ 97% on the pair subset alone, and strict gating aims to preserve baseline overall accuracy (multi-seed aggregates in the supplement / released JSON).

## 5 Limitations (domain shift)

- **No Indian phone-on-bag corpus yet.** All numbers use IRRI contact sensors plus ambient clean windows. The robustness ladder is a proxy, not a substitute for field labels.
- **Species coverage.** Pulse beetle and other pests are out of scope.
- **Reference comparison** to 84.51% is soft (protocol-matched claim, not identical reimplementation).
- **Single geography / grain type** in source data; storage bag materials and vernacular mics vary.

- **Screening aid only:** not legal or lab diagnosis; low-confidence UX must abstain.

## 6 Open release and outlook

Apache-2.0 code, distilled weights, bake-off tables, robustness/calibration JSON, and a static results dashboard are public.<sup>1</sup> Next: when feasible, phone-mic field collection with extension partners, and tighter phone-matched augmentation. Ground-up, small-model, offline tooling matches CCAI's call for resource-aware climate ML.

## References

C. Balingbing et al. Application of a multi-layer CNN and acoustic device for stored rice pests. *Computers and Electronics in Agriculture*, 225:109297, 2024.

IGMRI. Annual Report. Indian Grain Storage Management & Research Institute, 2015.

World Bank / FAO reports on post-harvest loss (food loss & waste). See also regional grain storage extension literature.

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<sup>1</sup>Anonymized for review; camera-ready will cite the repository.